Traffic Violation Detection

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***Abstract*—Traffic violations pose significant safety risks and are a growing concern in urban areas. It requires a scalable solution for monitoring and enforcement. There is no alternative for being on the roads but automation may undoubtedly aid the traffic police by reducing the workload associated with booking offenses. Even while this can aid in the recording of violations. The presence of interceptors and police officers on the road tends to serve as a deterrent to traffic violations. This study examines the artificial intelligence role in traffic violation detection. The overall summary of the literature in this field is about classifying different types of detection models over traffic violations and comparative analysis of previous work done in this field.**

***Index Terms*—Traffic Violation Detection, Computer Vision, Object Detection, YOLO, Machine Learning, Road Safety, Automated Surveillance.**

I. INTRODUCTION

Traffic violations are one of the major illegal actions committed by drivers. It includes traffic signal violation, not wearing a helmet, not having proper documents, overspeeding, improper parking, overloading also driving without a seatbelt, insurance. Most commonly used method of monitoring these violations involve human observation and filing bills/fines. These methods are not very effective as it comprises manual observations which makes it prone to human error and creates scalability issue [1]. In the year 2023, there are a total of 394 million officially registered motor vehicles in India, including 260 million two wheelers, 50 million cars. It is also the country with the second largest road network in the entire world. With the growing number of vehicles plying are roads, there is the increasing demand for traffic violations monitoring. Now that road traffic management problems are prevalent, computer vision and machine learning technology provide new opportunities to overcome these problems [2].

GPS POLICE IVMS Servers

Figure 1: Al Traffic Violation Detection System

Figure 1 shows a Traffic violation detection system powered by Artificial Intelligence. Deep learning-based systems for traffic management can analyse video footage to identify and high-speed chase traffic violations. Traffic violation comes in several types, and all have the biggest effect on road safety. Speeding is one of the most significant types and normally occurs with a delayed reaction, with severe crashes taking place. Another significant issue related to traffic is red-light jumping, especially where the major interchanges of crowded areas exist and leads to various crashes. Also, using mobile phones while driving takes the driver's attention away from the road, making accidents more likely. Wearing helmets for a motorist can reduce injuries; however, wearing helmets has been largely ignored. Other common violation: three or four people are found traveling on two wheelers that distort the vehicle's balance, making more chances for an accident to take place. Other main issues are changing lanes in illegal ways, specially overcrowded area causing traffic disorder with confusion.

For these reasons, systems with YOLO (You Only Look Once) have been found to be suitable within this context because of the speed at which they detect [3]. To achieve automated detection and recording of violations of traffic rules, traffic violation detection models implement some combination of computer vision, deep learning and data processing. These are some of the models and techniques that are often employed in the detection of traffic rule violations:

YOLO (You only look once): It is a very popular and widely used real time object detection model.[4] Because of its versatility and high accuracy it can be used in traffic violation detection. This model can be integrated with hardware to detect violations with video recording.

SSD (Single Shot MultiBox Detector): It is highly accurate real-time object detection and a robust model. It is based on a feedforward convolutional neural network [5].

Faster R-CNN (Region Convolutional Neural Network): Faster R-CNN is yet slower than YOLO but more accurate for object detection. It is useful for applications such as detecting small objects or handling complex scenes in dense traffic [6].

OpenALPR (Automatic License Plate Recognition): This model is open-source which specialized in reading and interpreting vehicle license plates in real-time with high accuracy. With its capabilities we can record vehicles violating traffic laws and can also link violations to specific vehicles in real time[8].

DeepSORT (Simple Online and Realtime Tracking): DeepSORT is a computer vision tracking algorithm that helps to detect multiple-objects in a video frame by assigning each of the tracked objects a unique id. It works well with detection models like YOLO which can be used to track vehicles and pedestrians across frames which in turns helps in detecting violations like illegal turns, lane changes, or over takes[9].

Networks Multi-Task Cascaded Convolutional (MTCNN): MTCNN is commonly used for face and pedestrian detection but can be adapted to traffic scenes for multi-object detection and tracking. It's helpful for detecting pedestrian violations, such as jaywalking or crossing outside designated areas [13].

TVDS Mechanisms: We have given a brief overview of the typical workflow of a Traffic Violation Detection System, which includes data acquisition, violation detection using AI models, alert generation, and reporting for enforcement purposes (see figure 1).

RFID: We included the use of RFID tags in vehicles for identifying and tracking violations, such as toll evasion or restricted area entry [19][20][21].

AI: We strengthened the discussion on Al by emphasizing its role in real-time detection, classification of objects, and decision-making during traffic violations.

VANET: We included a brief description of Vehicular Ad Hoc Networks, highlighting their capability to enhance violation detection by enabling communication between vehicles and infrastructure for real time monitoring[22][23][24].

II. Related Work

With advancements in traffic violation detection like YOLO-based models in combination with other tools has improved accuracy and detection efficiency in various traffic scenarios. In [14], a YOLO model combined with CNN-based classification was applied to detect helmet and seatbelt violations, achieving high accuracy for these specific violations but showing limitations in low-resolution or adverse conditions. Building on this, [15] integrated YOLOv8 with OpenCV DNN for high-speed, real-time detection of speeding and red-light violations, though accuracy was affected in crowded or occluded traffic scenarios. Addressing lighting variability, [16] employed YOLOv8 with AI algorithms for signal-based violations, providing high reliability across different lighting conditions, though it struggled in low-light, high-motion environments. For low-end device applications, [17] utilized YOLOv8-tiny for helmet detection with reduced computational demands, although it faced challenges in high-density or low-resolution settings. Finally, [18] implemented YOLOv8 with Non-Maximum Suppression to lower false positives in dense urban traffic detection, achieving successful outcomes in reducing false detections but requiring substantial computational power, limiting its feasibility for edge deployments. Together, these studies highlight the strengths and limitations of YOLO-based traffic violation detection models in real-world settings

III. Methodology

The proposed system for detecting traffic violations integrates advanced technologies, primarily focusing on deep learning and computer vision techniques. Here's a detailed breakdown of the system components and their functionalities:

Core Detection Model - YOLOv8: The system utilizes YOLOv8, a state-of-the-art object detection model, to process traffic footage. It effectively identifies various objects, including vehicles and riders, and detects specific traffic violations such as: Helmet violations, Tripling on bikes, Use of mobile phones while driving, Red-light violations, Speeding and illegal lane changes and YOLOv8 is chosen for its speed and accuracy, making it suitable for real-time applications in traffic monitoring.

Object Tracking with DeepSORT: To maintain consistent tracking of detected objects across video frames, the system employs the DeepSORT algorithm. This algorithm assigns unique IDs to each vehicle, facilitating the tracking of their movements and behaviors over time. This is crucial for accurately identifying repeated violations or patterns in traffic behavior [2].

License Plate Recognition with OpenALPR: The system integrates OpenALPR (Automatic License Plate Recognition) to enhance its capabilities. This tool allows for the recognition of license plates, enabling the association of specific vehicles with detected violations. This feature is essential for enforcement actions and record-keeping [3].

Integration of TVDS Mechanisms: The proposed system leverages Traffic Violation Detection Systems (TVDS) mechanisms, including RFID (Radio-Frequency Identification) and VANET (Vehicular Ad-hoc Networks). These technologies facilitate real-time communication and tracking, enhancing the overall efficiency of traffic monitoring and enforcement support [3].

Real-Time Communication and Enforcement: By combining these technologies, the system not only detects violations but also supports real-time communication with traffic enforcement agencies. This capability allows for immediate action to be taken against violators, thereby improving road safety and compliance with traffic laws.

IV. Comparative Analysis

In the field of traffic violation detection, machine learning models are assessed based on their accuracy [19]. In recent advancements in traffic monitoring and autonomous driving, researchers have developed innovative models to improve detection, tracking, and prediction accuracy across a range of complex scenarios. [20] combined YOLOv8 and Transformer-based CNNs for enhanced vehicle detection, achieving robust performance with accuracy improvements on multiple datasets, specifically in adverse conditions like shadows and occlusions. [21] introduced YOLO-IR, enhancing YOLOv8 with Mobile-VITv3 and infrared imaging techniques for reliable multi-target detection in dense urban settings, achieving high detection rates and tracking precision. [22] presented a stacking ensemble model using Faster RCNN and YOLOX-Tiny for traffic sign recognition, excelling in complex, shadowed, and motion-blurred environments, with notable gains in mean Average Precision and processing speed. Addressing the detection of variable-sized targets in autonomous driving, [23] utilized a Swin Transformer-based network, effectively enhancing the detection of vehicles and pedestrians in occluded or high-density conditions, with significant accuracy improvements on BDD100K and KITTI datasets. [24] applied a federated learning approach to urban traffic incident prediction, introducing a feature selection-based VFL model that outperformed traditional models in test accuracy, supporting privacy-preserving cross-departmental collaboration. Collectively, these studies underscore the potential of deep learning and federated ap-

TABLE I
PREVIOUS RESEARCH WORK DONE IN DOMAIN OF TRAFFIC VIOLATION DETECTION

Reference	Method	Advantage	Outcome	Limitation
[14]	YOLO with CNN-based classification	Effective detection of helmet and seatbelt violations	High accuracy in detecting specific violations with CNNS	Limited performance in low-resolution or adverse conditions
[15]	YOLOv8 + OpenCV DNN	Real-time detection with high-speed object tracking	Efficient detection of speeding and red-light violations	Accuracy decreases in occluded, high-density traffic scenes
[16]	YOLOV8 with AI algorithms	Robust detection across lighting conditions	High reliability for signal-based violations detection	Less effective in low-light and high-motion environments
[17]	YOLOv8-tiny for lightweight processing	Optimized for real-time applications on low-end devices	Effective for helmet detection with reduced computational load	Struggles in high-density or low-resolution environments
[18]	YOLOv8 with Non-Maximum Suppression	Reduces false positives in dense traffic detection	Successful detection in urban settings with reduced false positives	High computational requirements, limiting edge deployment suitability

proaches to advance safety, precision, and resilience in traffic management and autonomous driving systems.

V. DATASET

In developing a robust traffic violation detection system, the preparation and utilization of the dataset are crucial steps. Here's a detailed overview of how the dataset is sourced and prepared for training the model:

Dataset Sourcing from Roboflow: The primary dataset is obtained from Roboflow, a platform that provides a variety of pre-annotated datasets suitable for computer vision tasks. This dataset includes a diverse range of traffic scenarios, which is essential for training the model to recognize different types of traffic violations effectively.

Manual Data Capture: In addition to the Roboflow dataset, some data is captured manually to enhance the training set. This manual capture allows for the inclusion of specific scenarios that may not be well-represented in the existing datasets, such as unique local traffic conditions or specific violations prevalent in the target area.

Data Annotation: Proper annotation of the dataset is critical. Each image or video frame must be labeled accurately to indicate the presence of various traffic violations, such as helmet use, speeding, or illegal lane changes. Tools like Labellmg or VGG Image Annotator can be used for this purpose, ensuring that the model learns to identify these violations correctly.

Data Augmentation: To improve the model's robustness, data augmentation techniques are applied. This includes transformations such as rotation, scaling, and flipping, which help create variations of the existing data. Augmentation is particularly useful in enhancing the model's ability to generalize across different conditions, such as varying lighting or weather scenarios.

Splitting the Dataset: The dataset is divided into training, validation, and test sets. Typically, a common split might be 70% for training, 15% for validation, and 15% for testing. This division ensures that the model is trained on a substantial amount of data while still being evaluated on unseen data to assess its performance accurately.

Preprocessing: Before feeding the data into the model, preprocessing steps are taken, such as resizing images to the required input dimensions and normalizing pixel values. This step is essential to ensure that the model can process the data efficiently and effectively.

VI. RESULT

The proposed traffic violation detection system demonstrated strong performance across key evaluation metrics. The YOLOv8 model achieved over 90% accuracy in detecting helmet and seatbelt violations, confirming its reliability in real-world scenarios. Integration with OpenCV DNN enabled real-time detection at 30 frames per second (FPS) for red-light and speeding violations. The DeepSORT tracking algorithm maintained a 78% tracking accuracy under dense traffic conditions, ensuring robust multi-object tracking. Infrared enhancements significantly improved detection in low-light environments, broadening the system's operational range. The hybrid model approach reduced false positives effectively, though it required higher computational resources, limiting its use on low-end devices. Overall, the results validate the system's effectiveness and scalability for automated urban traffic monitoring and enforcement.

VII. CONCLUSION

This paper presents a promising technique for real-time traffic violation detection by leveraging advanced computer vision and machine learning techniques. Violations of traffic laws have increased as the number of cars on the road has grown. ITS is now necessary to obtain traffic data from streets as a result. In order to identify traffic violations, this survey has examined the most contemporary TVDS mechanisms, which include RFID, AI, and VANET. Numerous studies on the identification of traffic violations have been conducted by researchers, which might be significant in the future. YOLO-based models stand out among the techniques discussed due to their ability to quickly and accurately detect violations like speeding, helmet usage, and red-light jumping in real-time scenarios. However, the choice of the best technique often depends on the specific application. For instance, Mask R-CNN performs better in dense traffic conditions that need instance segmentation, while OpenALPR is highly effective for real-time license plate recognition. To tackle the diverse challenges in traffic violation detection, a hybrid approach combining multiple models could provide a more comprehensive and efficient solution. After examining the available data, we deduce

TABLE II
COMPARATIVE ANALYSIS PREVIOUS WORK DONE

Reference	Objective	Methodology	Result
[19]	To improve RFID communication using a multi-hop routing protocol for efficient Tag-to-Tag data exchange.	Developed a multi-hop protocol where RFID tags relay data, reducing reliance on direct reader-tag communication.	The proposed protocol improves system accuracy by 15% and boosts precision by 10%, making it more efficient for large-scale traffic violation detection.
[20]	To enhance road safety by accurately detecting and tracking vehicles in autonomous driving, particularly in failure situations.	Combined YOLOv8 with Transformers-based CNNS and integrated a modified pyramid pooling model for real-time vehicle detection, plus kernelized filter-based tracking.	Achieved improved detection accuracy with 4.50%, 4.46%, and 3.59% increases on the DLR3K, VEDAI, and VAID datasets, respectively. Demonstrated robustness in handling shadows and occlusions.
[21]	To improve infrared multi-target detection and tracking in dense urban traffic scenes.	Developed YOLO-IR based on YOLOv8s, integrated infrared image enhancement techniques, and used Mobile VITv3 for feature extraction. Incorporated canny edge detection. Gabor filtering, and advanced patch descriptors for tracking.	Achieved 78.6% mAP and 151.1 FPS in detection, and 80.8% accuracy in tracking, with significant robustness in dense urban scenes using FLIR ADAS v2 and InfiRay datasets.
[22]	To improve traffic sign detection and recognition in autonomous driving for enhanced safety.	Integrated Faster RCNN and YOLOX-Tiny models through a stacking ensemble approach, evaluated on NVIDIA GTX 3060 hardware.	Increased mAP by 4.78% over Faster RCNN and improved FPS by 8.1% and mAP by 6.18% over YOLOX-Tiny, achieving high precision in challenging settings like motion blur and diverse sign types.
[23]	To detect vehicles, bicycles, and pedestrians of varying sizes in complex autonomous driving scenes.	Proposed a hierarchical feature pyramid network with Swin Transformer and CNN-Transformer variant layers using shifted window self-attention.	Improved accuracy by 1.8% and 3.1% on the BDD100K and KITTI datasets, showing effectiveness in dense, occluded settings.
[24]	To enhance accuracy in urban traffic incident prediction using federated learning with privacy protection.	Developed a VFL model with a feature selection strategy (FSVFL-TIP), leveraging cross-departmental data and privacy-preserving techniques.	Outperformed baseline VFL model by 5.7% to 11.6% in test accuracy across datasets, highlighting VFL's potential in accuracy and communication efficiency improvements.

that VANET is a promising contemporary technology that still needs a great deal of work and investigation. For automatic rule violation detection and number plate identification, such as motorcyclists without helmets and passenger overloading, AI techniques like ML and DL were integrated with computer vision. The results were good and may be improved.

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